

kinematics - practice problems

name _____

1. A dynamics cart rolls along a 2.0 m track in physics class. A motion sensor samples its position 50 times each second. At 0.40 s the cart is at the 35 cm mark, and at 2.40 s it is at the 155 cm mark. What is its average velocity?
2. A sprinter in a 400 m race leaves the starting blocks from rest and reaches 9.0 m/s in 3.0 s. What is the sprinter's average acceleration?
3. A skateboarder is rolling at 18 km/h when they start down a gentle hill. The hill gives them a constant acceleration of 0.50 m/s^2 for 6.0 s. How fast are they moving at the bottom?
4. A friend gives a sled a push so it starts down a snowy hill at 2.0 m/s. The rider is wearing a 4 kg backpack. The sled speeds up with a constant acceleration of 0.80 m/s^2 for 10 s. How far does the sled travel down the hill?
5. A car speeds up at a constant rate along a 150 m highway on-ramp. At the end of the ramp it is moving 27 m/s. How long is the car on the ramp?

6. A motorcycle is already moving at 12 m/s when it enters a 50 m straight section of road. It accelerates at 2.0 m/s^2 along that section. How fast is it moving at the end?
7. A car is moving at 100 km/h when the driver slams on the brakes. The car slows down at -7.0 m/s^2 until it stops. How far does the car travel while braking?
8. A car leaving a school zone speeds up at a constant rate. It covers 90 m in 6.0 s, and at the end it is moving 18 m/s. How fast was it moving at the start?
9. You drop a rock from rest into a canyon. You hear it hit the bottom 3.0 s later. About how deep is the canyon?
10. To serve, a volleyball player tosses the ball straight up at 5.0 m/s. How high above the player's hand does the ball rise?
11. A batter pops a baseball straight up at 24 m/s. How long does it take the ball to reach its highest point?

12. A stone is thrown straight down at 5.0 m/s from the top of a 30 m cliff. How fast is it moving just before it hits the ground, and how long does the fall take?
13. Two identical balls are at the same height. At the same moment, one is dropped straight down and the other is fired horizontally at 50 m/s. Ignoring air friction, which ball hits the ground first?
14. A mountain biker rides off a 120 cm high ledge while moving horizontally at 5.0 m/s. How long is the biker in the air, and how far from the base of the ledge do they land?
15. A marble rolls off the edge of a 0.80 m tall table and lands on the floor 0.60 m from the base of the table. How fast was the marble rolling when it left the table?
16. A ball is launched at 25 m/s at an angle of 37° above the horizontal. What are its initial horizontal and vertical velocity components?
17. A ball is launched from level ground at 72 km/h and 30° above the horizontal. How long is it in the air, and what is its range?

18. A high school shot putter releases the shot 2.1 m above the ground at 13 m/s and 40° above the horizontal. How far forward does the shot travel before it hits the ground?

19. A water balloon is thrown from a balcony 4.0 m above the ground at 18 m/s and 25° above the horizontal. How fast is it moving when it hits the ground?

20. A launcher gives a ball a horizontal velocity of 12 m/s. A target opening is 18 m away and 4.0 m above the launch height. What initial vertical velocity does the ball need to pass through the opening?

21. A soccer player takes a free kick. The ball leaves the ground with a horizontal velocity of 20 m/s and a vertical velocity of 6.0 m/s. The defenders' wall is 9.15 m away, and the players can jump to block anything below 2.2 m. Does the ball make it over the wall?

22. A cyclist's bike computer records these speeds while the cyclist rides in a straight line. Is the acceleration constant? If it is, find the acceleration and how far the cyclist travels in these 3.0 s.

time (s)	0	1.0	2.0	3.0
velocity (m/s)	2.0	4.5	7.0	9.5